

Junior Topology Talk - Studying Concordance using Instantons

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10/29/2025

Abstract

Last week, Hillary told us about the relationship between satellite knots and concordance groups, mentioning a conjecture of Hedden stating that satellite operations almost never induce homomorphisms. In this talk, I'll investigate a slightly different conjecture regarding the *rank* of satellite maps (also due to Hedden). Using a clever application of Chern-Simons gauge theory, we'll show that a large class of satellite maps have infinite rank. In special situations we'll get even more, satellite maps which are *zero* on the topological concordance group (i.e. has image consisting of topologically slice knots) but are infinite rank on the smooth concordance group. No gauge theory background is required! But I will make brief comparisons with Seiberg-Witten theory.

1 Intro

Let $\mathcal{C}$ be the group of knots up to smooth concordance in S^3 . Given a pattern P , recall that the *winding number* of P is the absolute value of $[P] \in H_1(S^1 \times D^2) \cong \mathbb{Z}$.

We'll focus on this main conjecture:

Conjecture 1. *The image of every non-constant satellite operator has infinite rank.*

This turns out to be true for a large number of satellite operators (hence the conjecture). This talk will focus on proving the following:

Theorem 2. *(Hedden-Pinzón-Caicedo '21) Let $P \subset S^1 \times D^2$ be a pattern with winding number zero. If ∂D^2 in S^3 has a lift to $\Sigma_2(P(U))$ which links itself non-trivially, then $P : \mathcal{C} \rightarrow \mathcal{C}$ has infinite rank.*

Obvious questions following the statement of this theorem include the following: **Q:** Why winding number zero? **A:** It's actually straightforward to prove this result for winding number non-zero patterns, I'll try to do so if there's time.

Q: Why would we use gauge theory for this problem? Surely there are other concordance invariants which can detect infinite rank. Indeed, this is not the case. **A:** First, note that that this theorem is *inherently* a smooth theorem — any pattern which is unknotted in S^3 acts by zero on the topological concordance group. It is very difficult, maybe impossible, to prove this result using other known smooth concordance invariants. Knot Floer homology, Khovanov homology, or generalizations thereof are difficult to compute for large families of knots, and are not expected to behave predictably under satellites.

However, work of Fintushel-Stern '85, Furuta '90, and later refinements by Hedden-Kirk '11,'12, has demonstrated that instantons yield effective techniques to prove these independence results, in particular by obstructing the existence of certain cobordisms. This talk is about a recent application of these techniques.

Q: Okay, so why not use Seiberg-Witten or some other gauge theory? **A:** It turns out that, despite the enormous progress in other gauge theories, no alternative proof of these cobordism obstructions have been found. Instantons are the only way we know how to do it!

2 Case of non-zero winding number (Maybe)

Now we'll look at the case of winding number = 0.

3 From concordance to $\mathbb{Z}/2$ -homology cobordism

Definition 3. The **3-dimensional $\mathbb{Z}/2$ homology cobordism group**, denoted $\Theta_3^{\mathbb{Z}/2}$, is the group of $\mathbb{Z}/2$ -homology 3-spheres up to $\mathbb{Z}/2$ -homology cylinders under the operation of connected sum.

How is this related to concordance? There is a map

$$\Sigma_2 : \mathcal{C} \rightarrow \Theta_3^{\mathbb{Z}/2}$$

given by sending a concordance class of a knot K to its 2-fold branched cover $\Sigma_2(K)$.

Proposition 4. *This map is well defined and a homomorphism.*

Proof. Let K_1, K_2 be knots in S^3 . Suppose that K_1 and K_2 are concordant via the surface $C \subset S^3 \times I$. Then $\Sigma_2(C)$ is a $\mathbb{Z}/2$ -homology cylinder cobounding $\Sigma_2(K_1)$ and $\Sigma_2(K_2)$. Hence Σ_2 is well defined. Now, observe that

$$\Sigma_2(K_1 \# K_2) \sim \Sigma_2(K_1) \# \Sigma_2(K_2).$$

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□

Thus, if we want to prove Theorem 2, it suffices to find an infinite collection of knots $\{K_n\}_{n=1}^{\infty}$ such that $\{\Sigma_2(P(K_n))\}_{n=1}^{\infty}$ are linearly independent in $\Theta_3^{\mathbb{Z}/2}$.

4 An Instanton Obstruction to Negative Definite Cobordism

4.1 Instanton crash course

I assume most members of the audience, even those with some exposure to gauge theory, haven't seen instantons before. I'll give a very brief and broad overview, and try to compare with Seiberg-Witten theory. Ultimately, the theory of instantons is much more natural than Seiberg-Witten theory (at least for me). Instanton Floer homology was Floer's original theory which spurred so many spin offs (anything with Floer in the name). As usual, there are distinct stories for 3-manifolds and for 4-manifolds. In the latter case, if X is a 4-manifold and $E \rightarrow X$ is an $\mathrm{SO}(3)$ - or $\mathrm{SU}(2)$ -bundle, one studies the moduli space $\mathcal{M}$ of anti-self-dual (ASD) connections (*instantons*) on E modulo gauge transformation. The analog of the Seiberg-Witten equations are the ASD equations, and the two theories operate under very similar analytic machinery. It turns out that $\mathcal{M}$ is generically a finite-dimensional, orientable manifold — although, unlike the Seiberg-Witten situation, it is *not* usually compact. However, the failure for $\mathcal{M}$ to be compact is well understood. This will be key to our argument.

This story generalizes to *adapted* bundles over 4-manifolds with *tubular ends*. Now, the input data is a 4-manifold X with has a finite number of “ends”, open subsets U_i of X isometric to $Y \times (0, \infty)$, and an $\mathrm{SO}(3)$ - or $\mathrm{SU}(2)$ -bundle $E \rightarrow X$ with a fixed flat connection on each end. One should think of these fixed connctions as the “limiting” connection along that end. So to any adapted bundle $(E, \underline{\alpha})$ where $\underline{\alpha} = (\alpha_1, \dots, \alpha_n)$ are our fixed end connections, we obtain an instanton moduli space $\mathcal{M}(E, \underline{\alpha})$.

¹TODO: picture

4.2 Compactness

As in the closed 4-manifold case, compactness is well understood. There is an invariant associated with each end of a 4-manifold with tubular ends which is sufficient to guarantee compactness of $\mathcal{M}(E, \underline{\alpha})$. There are essentially two types of ways $\mathcal{M}(E, \underline{\alpha})$ can be non-compact: by energy bubbling off interior points and by energy escaping down the ends of X . This is the *convergence with no loss of energy* theorem of Morgan-Mrowka-Ruberman. Given an end (Y_i, α_i) , we can associate to it a number $\tau(Y_i, \alpha_i) \in (0, 4]$ which provides a lower bound on the amount of energy that must escape over the end, if any escapes at all. Informally, $\tau(Y_i, \alpha_i)$ is the minimal relative Chern-Simons invariant $cs(Y_i, \alpha_i, \gamma)$ where γ ranges over flat connections on $E|_{Y_i}$, taken modulo 4.

Proposition 5. *Suppose that $(E, \underline{\alpha})$ is an adapted bundle with each α_i non-degenerate and*

$$0 \leq p_1(E, \alpha) < \min \tau(Y_i, \alpha_i).$$

Then $\mathcal{M}(E, \underline{\alpha})$ is compact.

The “taken mod 4” is necessary to eliminate the possibility of bubbling, since energy bubbles off interior points in multiples of 4.

4.3 Obstructing Cobordisms

So, how can we use this to demonstrate linear independence in $\Theta_3^{\mathbb{Z}/2}$? Here is our main obstruction result:

Theorem 6. *(Furuta '90 ($p = 1$), Hedden-Kirk '11 ($p > 1$)) Let p, q and r, s be two pairs of relatively prime, positive integers, and suppose $\{\Sigma_i\}_{i=1}^N$ is a family of oriented $\mathbb{Z}/2$ -homology 3-spheres, where $\Sigma_N = S_{p/q}^3(T_{r,s})$. If*

$$\frac{p}{rs(qrs - p)} < \min \left\{ \frac{1}{r}, \frac{1}{s}, \frac{1}{qrs - p}, \tau(\pm \Sigma_1), \dots, \tau(\pm \Sigma_{N-1}) \right\},$$

then there does not exist a smooth, negative definite 4-manifold X with $H^1(X; \mathbb{Z}/2) = 0$ and whose boundary is given by

$$\partial X = \coprod_{i=1}^N a_i \Sigma_i,$$

where $a_i \in \mathbb{Z}, a_N > 0$.

Proof. (Sketch) The proof is by contradiction. The manifold $-\Sigma_N = -S_{p/q}^3(T_{r,s})$ is diffeomorphic to the Brieskorn sphere $\Sigma(r, s, qrs - p)$, which admits a description as the link of the complex surface singularity

$$z_0^r + z_1^s + z_2^{qrs-p} = 0.$$

In particular, $-\Sigma_N$ is Seifert fibered. Thus $-\Sigma_N$ bounds two closely related negative definite 4-manifolds. The first is the canonical resolution R of the singularity by repeated blowups. The second is the manifold obtained from the mapping cylinder of the Seifert fibration $-\Sigma_N \xrightarrow{\pi} S^2$ by excising neighborhoods of the singularities coming from the singular fibers.²

So, W has three additional lens space boundary components, $L(r, b_0)$, $L(s, b_1)$, and $L(qrs - p, b_2)$, where the b_i are determined but not worth writing out. Now, we construct an adapted $\text{SO}(3)$ -bundle over W . Let $E \rightarrow X$ be the bundle $\mathcal{L} \oplus \epsilon$, where $\mathcal{L}$ is an $\text{SO}(2)$ -bundle over W such that $c_1(\mathcal{L}) \in H^2(W) \cong \mathbb{Z}$ generates. Take the trivial flat connection α_0 on $-\Sigma_N$, and non-trivial flat connections $\alpha_1, \alpha_2, \alpha_3$ on the lens spaces.

²TODO: Picture

Attaching tubular ends to W , we obtain an instanton moduli space $\mathcal{M}(E, \underline{\alpha})$. Because these end connections are non-degenerate, the Atiyah-Patodi-Singer index theorem and the Neumann-Zagier formula allows us to compute the virtual dimension of $\mathcal{M}(E, \underline{\alpha})$ to be 1. Moreover, Hodge theory allows us to show that there is a unique reducible ASD connection, hence $\mathcal{M}(E, \underline{\alpha})$ is non-empty. The slice theorem gives a description of the neighborhood of this singular point as a half-open interval $[0, \epsilon)$.

Suppose that X as in the theorem statement exists. Cap off one of the Σ_N boundary components with W and the other $a_N - 1$ components with R to get $\widehat{X}$ such that

$$\partial \widehat{X} = L(r, b_0) \amalg L(s, b_1) \amalg L(qrs - p, b_2) \coprod_{i=1}^{N-1} a_i \Sigma_i.$$

Extend E to $\widehat{X}$ by the trivial bundles over X and R to get a new moduli space $\widehat{\mathcal{M}}$. Now, we can again enumerate the singular points of our moduli space to understand its boundary. This is more subtle, but because the boundary components of X are $\mathbb{Z}/2$ -homology spheres, this quantity is determined by the intersection form on X . Because $H^1(x; \mathbb{Z}/2) = 0$, the order of the torsion subgroup of $H^2(X, \partial X)$ is odd, which implies that the number of singular points in $\widehat{\mathcal{M}}$ is odd. But since $\widehat{\mathcal{M}}$ is 1-dimensional, it must be non-compact, since compact 1-manifolds have an even number of boundary points.

On the other hand, the energy of a point on $\widehat{\mathcal{M}}$ is given by $\frac{q}{rs(qrs-p)}$ (determined by $c_1(\mathcal{L})^2$). Since this quantity is less than 4, bubbling cannot occur, and since we've exactly supposed that the condition of Theorem 5 holds, energy cannot escape down the ends of $\widehat{X}$. So $\widehat{\mathcal{M}}$ is compact, a contradiction. $\square$

This theorem can be used to obstruct the existence of certain negative definite cobordisms relevant to Theorem 2. But first, let's take a break to do some topology.

5 Topological Setup

From here on out, let $P \subset S^1 \times D^2$ be a winding number 0 pattern, (p, q) and (r, s) be pairs of coprime non-zero integers where p is odd, and K a knot in S^3 .

Let Σ be a rational homology 3-sphere containing two disjoint oriented closed curves γ and η . There is a natural $\mathbb{Q}$ -valued linking form determined by the isomorphism given by Alexander duality

$$H_1(\Sigma \setminus \gamma) \xrightarrow{\mathcal{A}_\gamma} \mathbb{Q} \langle \mu_\gamma \rangle,$$

i.e. $lk_\Sigma(\gamma, \eta)$ is given by $\mathcal{A}_\gamma(\eta) = lk_\Sigma(\gamma, \eta) \cdot [\mu_\gamma]$. It turns out that this form is symmetric. If a curve γ in Σ is framed, there is a well-defined self-linking number $lk_\Sigma(\gamma, \gamma)$.

Now, note that any lift of ∂D to a curve μ_K in $\Sigma_2(P(K))$ has a natural framing determined by the framing curve given by pushing off μ_K along the corresponding lift of $\partial(S^1 \times D^2) \cong S^1 \times S^1$. This framing is independent of K , and so we have a well-defined invariant ℓ_P of P given by self-linking number

$$\ell_P := lk_{\Sigma_2(P(U))}(\mu_U, \mu'_U).$$

Proposition 7. *If $\ell_P < -q/p$, then there exists a negative definite cobordism Z from $\Sigma_2(P(K))$ to $S^3_{p/q}(K) \# Y$ for some 3-manifold Y .*

Proof. Let η be a curve on $S^1 \times S^1$ representing the surgery slope $p\mu + q\lambda \in H_1(S^1 \times S^1)$. $\square$

Note however the following corollary to the instanton obstruction:

Corollary 8. *Let $\{\Sigma_i\}_{i=1}^\infty$ be a family of oriented $\mathbb{Z}/2$ -homology 3-spheres. Suppose there's a negative definite cobordism Z from Σ_N to $S_{p/q}^3(T_{r,s}) \# Y$ where $H^1(\mathbb{Z}; \mathbb{Z}/2) = 0$ and Y is a $\mathbb{Z}/2$ -homology 3-sphere. If*

$$\frac{p}{rs(qrs - p)} < \min \left\{ \frac{1}{r}, \frac{1}{s}, \frac{1}{qrs - p}, \tau(\pm \Sigma_1), \dots, \tau(\pm \Sigma_{N-1}), \tau(Y) \right\},$$

then Σ_N has infinite order in $\Theta_3^{\mathbb{Z}/2}$ and is independent from the other manifolds:

$$\langle \Sigma_N \rangle \cap \langle \Sigma_1, \dots, \Sigma_{N-1} \rangle = \{0\}.$$

Proof. By contradiction, suppose that this intersection is non-zero or that Σ_N has finite order. Then there's a $\mathbb{Z}/2$ -homology 4-ball Q with

$$\partial Q = c_N \Sigma_N \# \overline{(c_1 \Sigma_1 \dots \# c_{N-1} \Sigma_{N-1})}$$

and $c_N > 0$. Form the manifold

$$X' = Q \cup_{c_N \Sigma_N} \#_{c_N} Z.$$

Attach 3-handles along the connected sum spheres to get a smooth negative definite 4-manifold X with $H^1(X; \mathbb{Z}/2) = 0$ and

$$\partial X = c_N S_{p/q}^3(T_{r,s}) \amalg c_N Y \prod_{i=1}^{N-1} -c_i \Sigma_i.$$

This contradicts Theorem 6. □

6 Proof of Theorem 2

Now, our theorem looks like

Theorem. *If $\ell_P \neq 0$, there exists an infinite family of knots $\{K_i\}_{i=1}^\infty$ for which $\{P(K_i)\}_{i=1}^\infty$ is a $\mathbb{Z}$ -independent family in $\mathcal{C}$.*

Proof. Suppose first that $\ell_P \in \mathbb{Q}$ is strictly negative. Fix p, q with p odd and q having parity opposite that of the numerator of ℓ_P , and satisfying $\ell_P < -q/p < 0$. We recursively define the family $\{K_i\}_{i=1}^\infty$. Let $K_1 = T_{r_1, s_1}$ and $\Sigma_1 = \Sigma_2(P(K_1))$, where

$$\frac{p}{r_1 s_1 (q r_1 s_1 - p)} < \min \left\{ \tau(Y), \frac{1}{r_1}, \frac{1}{s_1}, \frac{1}{q r_1 s_1 - p} \right\}.$$

By Theorem 8, $\langle \Sigma_1 \rangle \cong \mathbb{Z}$.

In general, suppose that $K_1, \dots, K_{N-1}$ have been chosen such that

$$\langle \Sigma_1, \dots, \Sigma_{N-1} \rangle \cong \mathbb{Z}^{N-1}.$$

Choose r_N, s_N such that

$$\frac{p}{r_N s_N (q r_N s_N - p)} < \min \left\{ \tau(Y), \tau(\pm \Sigma_1), \dots, \tau(\pm \Sigma_{N-1}), \frac{1}{r_N}, \frac{1}{s_N}, \frac{1}{q r_N s_N - p} \right\}.$$

For $K_N = T_{r_N, s_N}$ and $\Sigma_N = \Sigma(P(K_N))$, Theorem 8 yields that

$$\langle \Sigma_1, \dots, \Sigma_N \rangle \cong \mathbb{Z}^N.$$

This shows that $\{P(K_i)\}_{i=1}^\infty$ is an independent family in $\mathcal{C}$.

Note that if ℓ_P is positive, then $\ell_{m(P)}$ is negative, where $m(P)$ is the mirror of P . Thus, if $\{m(P)(K_i)\}_{i=1}^\infty$ is independent, so too is $\{P(m(K_i))\}_{i=1}^\infty$. $\square$